

Customer Segmentation with Clustering

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import scipy.cluster.hierarchy as sch
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

sns.set_style('whitegrid')
```

```
!git clone 'https://github.com/HarshvardhanSingh-13/Datasets'
```

```
Cloning into 'Datasets'...
remote: Enumerating objects: 335, done.
remote: Counting objects: 100% (93/93), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 335 (delta 22), reused 0 (delta 0), pack-reused 242 (from 1)
Receiving objects: 100% (335/335), 302.68 MiB | 14.81 MiB/s, done.
Resolving deltas: 100% (136/136), done.
Updating files: 100% (243/243), done.
```

```
df = pd.read_csv('/content/Datasets/Mall Dataset/Mall_Customers.csv')
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                   200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.drop('CustomerID',axis=1,inplace=True)
```

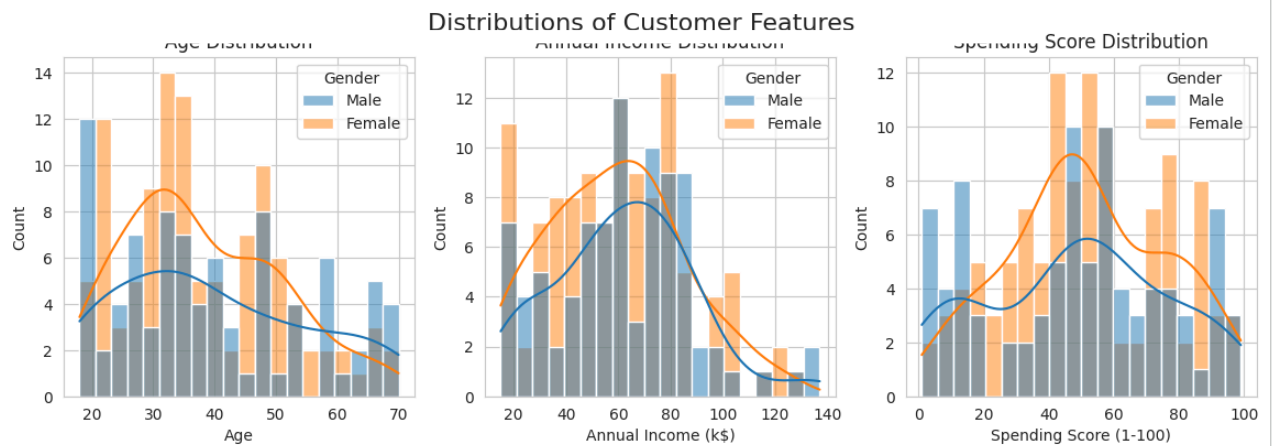
```
df.describe()
```

	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000
mean	38.850000	60.560000	50.200000
std	13.969007	26.264721	25.823522
min	18.000000	15.000000	1.000000
25%	28.750000	41.500000	34.750000
50%	36.000000	61.500000	50.000000
75%	49.000000	78.000000	73.000000
max	70.000000	137.000000	99.000000

```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
fig.suptitle('Distributions of Customer Features', fontsize=16)
```

```
sns.histplot(ax=axes[0], data=df, x='Age', kde=True, bins=20, hue='Gender').set_title('Age Distribution')
sns.histplot(ax=axes[1], data=df, x='Annual Income (k$)', kde=True, bins=20, hue='Gender').set_title('Annual Inc
sns.histplot(ax=axes[2], data=df, x='Spending Score (1-100)', kde=True, bins=20, hue='Gender').set_title('Spendi

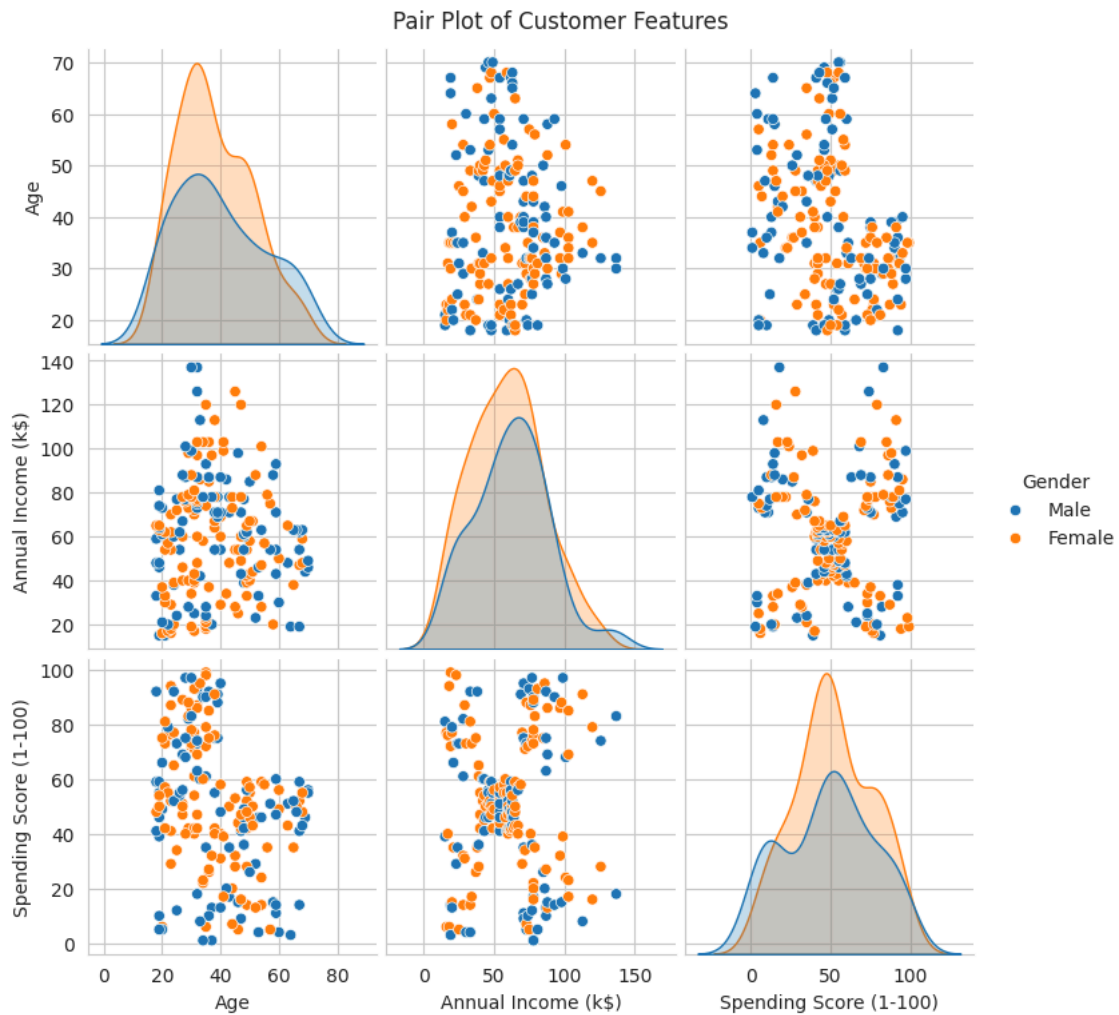
plt.show()
```



```
df.head(2)
```

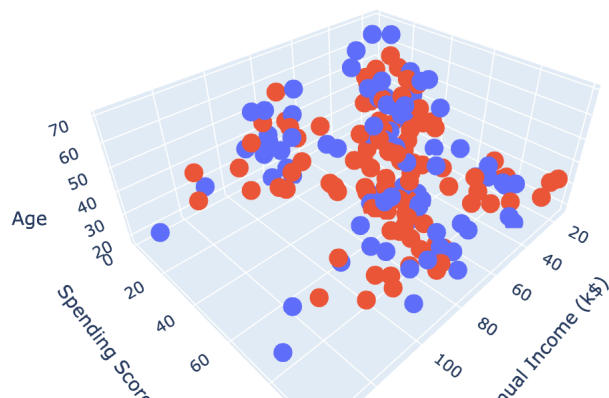
	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	Male	19	15	39
1	Male	21	15	81

```
sns.pairplot(df, vars=['Age','Annual Income (k$)','Spending Score (1-100)'], hue='Gender', diag_kind='kde')
plt.suptitle('Pair Plot of Customer Features', y=1.02)
plt.show()
```



```
fig = px.scatter_3d(
    df,
    x='Annual Income (k$)',
    y='Spending Score (1-100)',
    z='Age',
    color='Gender',
    title='3d Visual'
)
fig.show()
```

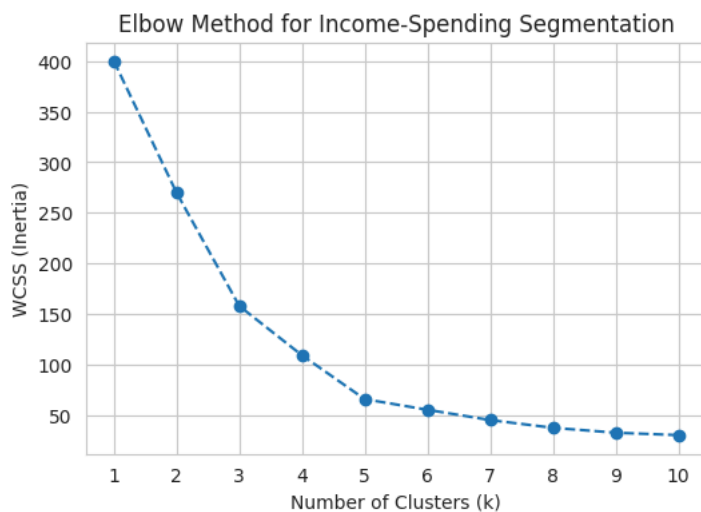
3d Visual



```
# 1. Select & Scale the values
X1 = df[['Annual Income (k$)', 'Spending Score (1-100)']]
scaler1 = StandardScaler()
X1_scaled = scaler1.fit_transform(X1)
```

```
# 2. Implement the Elbow Method
wcss1 = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X1_scaled)
    wcss1.append(kmeans.inertia_)
```

```
# 3. Plot the Elbow Curve
plt.figure(figsize=(6, 4))
plt.plot(range(1, 11), wcss1, marker='o', linestyle='--')
plt.title('Elbow Method for Income-Spending Segmentation')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS (Inertia)')
plt.xticks(range(1, 11))
plt.show()
```



```
kmeans1 = KMeans(n_clusters=5, init='k-means++', random_state=42, n_init=10)
df['Income_Cluster'] = kmeans1.fit_predict(X1_scaled)
```

```
df.head()
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income_Cluster
0	Male	19	15	39	4
1	Male	21	15	81	2
2	Female	20	16	6	4
3	Female	23	16	77	2
4	Female	31	17	40	4

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Annual Income (k$)', y='Spending Score (1-100)',
                hue='Income_Cluster', palette='viridis', s=100, alpha=0.8, edgecolor='black')
plt.title('Customer Segments by Income and Spending')
plt.legend(title='Cluster')
plt.show()
```



```
# Quantitative Persona Analysis
cluster_profiles1 = df.groupby('Income_Cluster')[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
cluster_profiles1['Size'] = df['Income_Cluster'].value_counts()
print("---- Income-Based Cluster Profiles ----")
cluster_profiles1
```

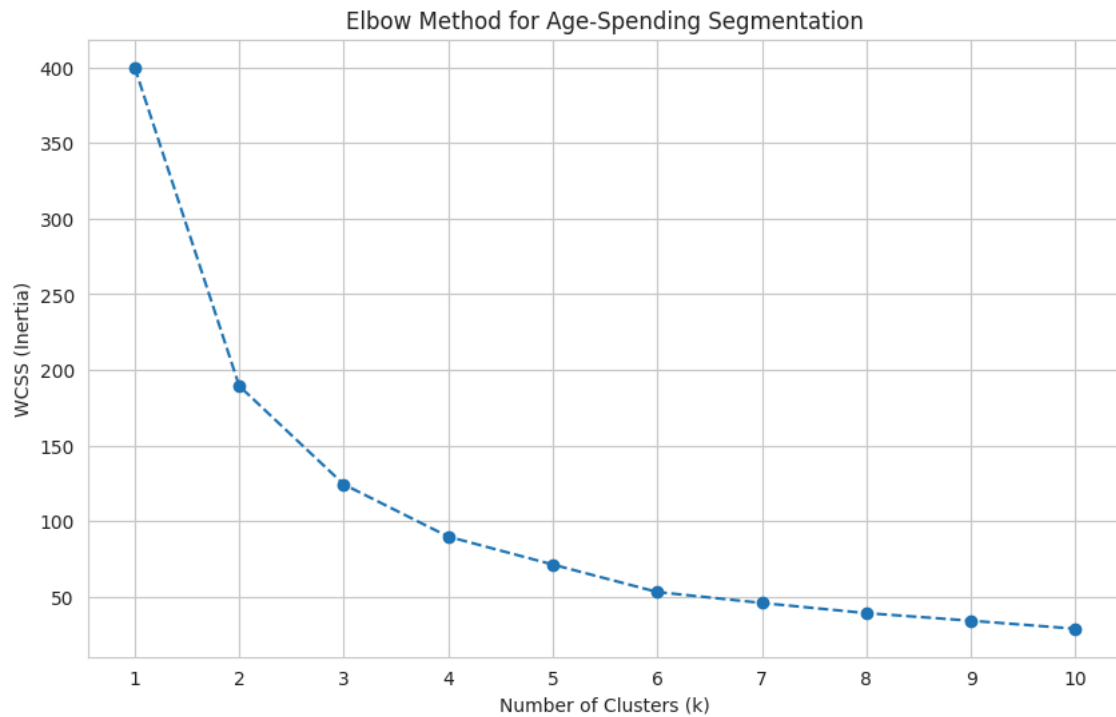
---- Income-Based Cluster Profiles ----

	Age	Annual Income (k\$)	Spending Score (1-100)	Size
Income_Cluster				
0	42.72	55.30	49.52	81
1	32.69	86.54	82.13	39
2	25.27	25.73	79.36	22
3	41.11	88.20	17.11	35
4	45.22	26.30	20.91	23

```
# 1. Select and scale the features
X2 = df[['Age', 'Spending Score (1-100)']]
scaler2 = StandardScaler()
X2_scaled = scaler2.fit_transform(X2)

# 2. Implement the Elbow Method
wcss2 = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X2_scaled)
    wcss2.append(kmeans.inertia_)

# 3. Plot the Elbow Curve
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss2, marker='o', linestyle='--')
plt.title('Elbow Method for Age-Spending Segmentation')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS (Inertia)')
plt.xticks(range(1, 11))
plt.show()
```

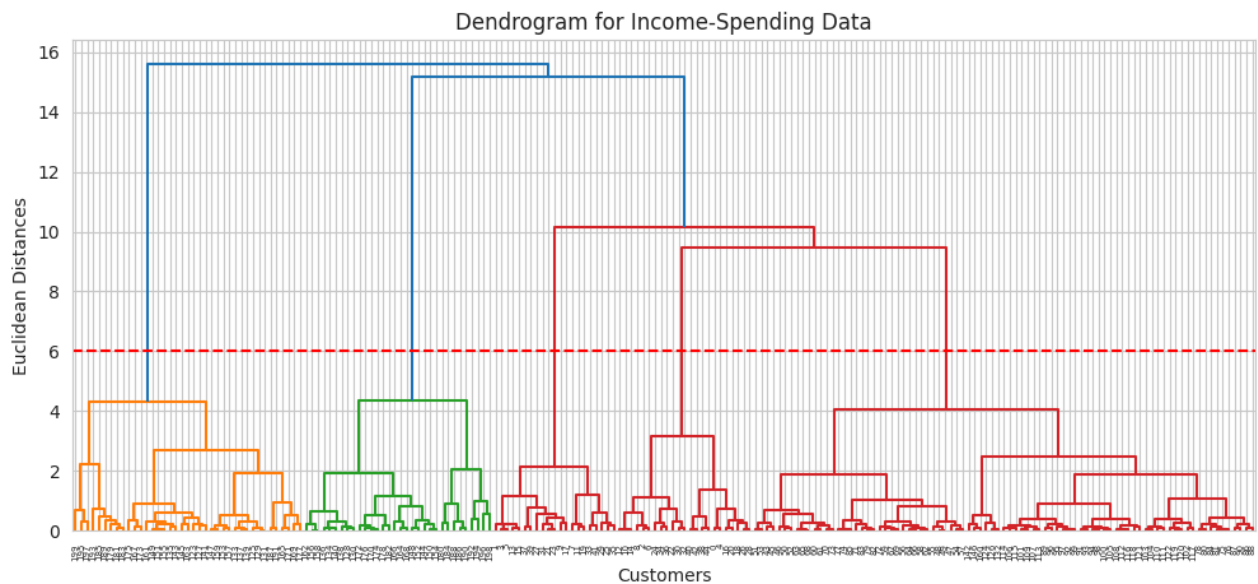


```
# Build and fit the final model for age segmentation
kmeans2 = KMeans(n_clusters=4, init='k-means++', random_state=42, n_init=10)
df['Age_Cluster'] = kmeans2.fit_predict(X2_scaled)

# Visualize the new clusters
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Age', y='Spending Score (1-100)',
                hue='Age_Cluster', palette='magma', s=100, alpha=0.8, edgecolor='black')
plt.title('Customer Segments by Age and Spending')
plt.legend(title='Cluster')
plt.show()
```



```
plt.figure(figsize=(12, 5))
dendrogram = sch.dendrogram(sch.linkage(X1_scaled, method='ward'))
plt.title('Dendrogram for Income-Spending Data')
plt.xlabel('Customers')
plt.ylabel('Euclidean Distances')
plt.axhline(y=6, color='r', linestyle='--') # Example cut-off line
plt.show()
```



Conclusion

In this extended project, we successfully applied unsupervised machine learning to perform customer segmentation.

Key Steps Undertaken:

1. **Performed an in-depth EDA:** Used 2D and 3D plots to explore customer data from multiple angles.
2. **Built two distinct segmentation models:** Created an income-based model and an age-based model, showing that segmentation is not a one-size-fits-all process.
3. **Used the Elbow Method** to programmatically determine the optimal number of clusters for each scenario.
4. **Introduced Hierarchical Clustering** as an alternative method and used its dendrogram to validate our choice of k .
5. **Created quantitative, data-driven personas** for each segment, providing the marketing team with precise, actionable insights for targeted campaigns.

This project highlights how different clustering approaches can uncover different facets of customer behavior, providing a richer, more nuanced understanding than a single model alone.

Submission

Submission Criteria Based on the project objectives and the insights gained from the initial analysis, here are the key criteria for your submission:

1. Explore Gender vs. Spending Score: Analyze the relationship between 'Gender' and 'Spending Score (1-100)' to see if gender plays a significant role in spending habits. Visualizations and summary statistics would be helpful here.
2. Apply Feature Engineering for Clustering: Create a new feature using existing data (Feature Engineering). Then, explore if clustering can be effectively performed using this newly engineered feature in combination with one or more existing features. Document the feature engineering process, the chosen features for clustering, the determination of the optimal number of clusters (e.g., using the Elbow Method), and the resulting cluster visualization and interpretation.

1. Explore Gender vs. Spending Score: Analyze the relationship between 'Gender' and 'Spending Score (1-100)' to see if gender plays a significant role in spending habits. Visualizations and summary statistics would be helpful here.

```
df['gender_binary'] = df['Gender'].map({'Male': 1, 'Female': 0})
print(df.head())
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income_Cluster	\
0	Male	19	15	39	4	
1	Male	21	15	81	2	
2	Female	20	16	6	4	
3	Female	23	16	77	2	
4	Female	31	17	40	4	

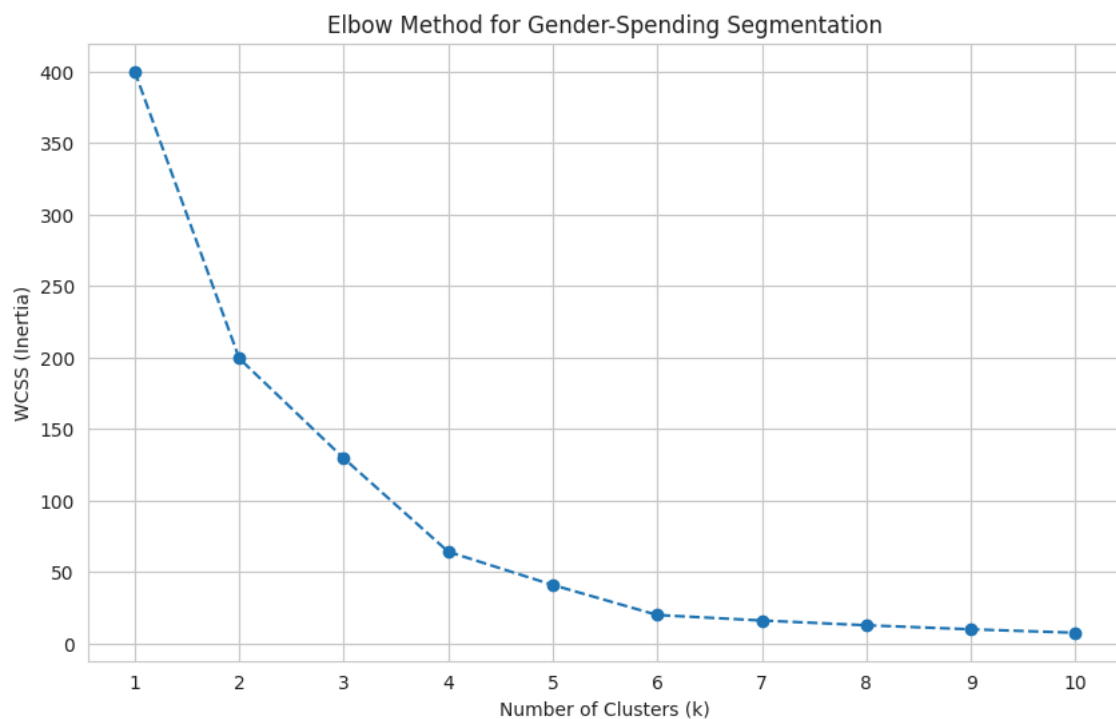
	Age_Cluster	gender_binary
0	0	1
1	1	1

2	0	0
3	1	0
4	0	0

```
# 1. Select and scale the features
X3 = df[['gender_binary', 'Spending Score (1-100)']]
scaler3 = StandardScaler()
X3_scaled = scaler3.fit_transform(X3)

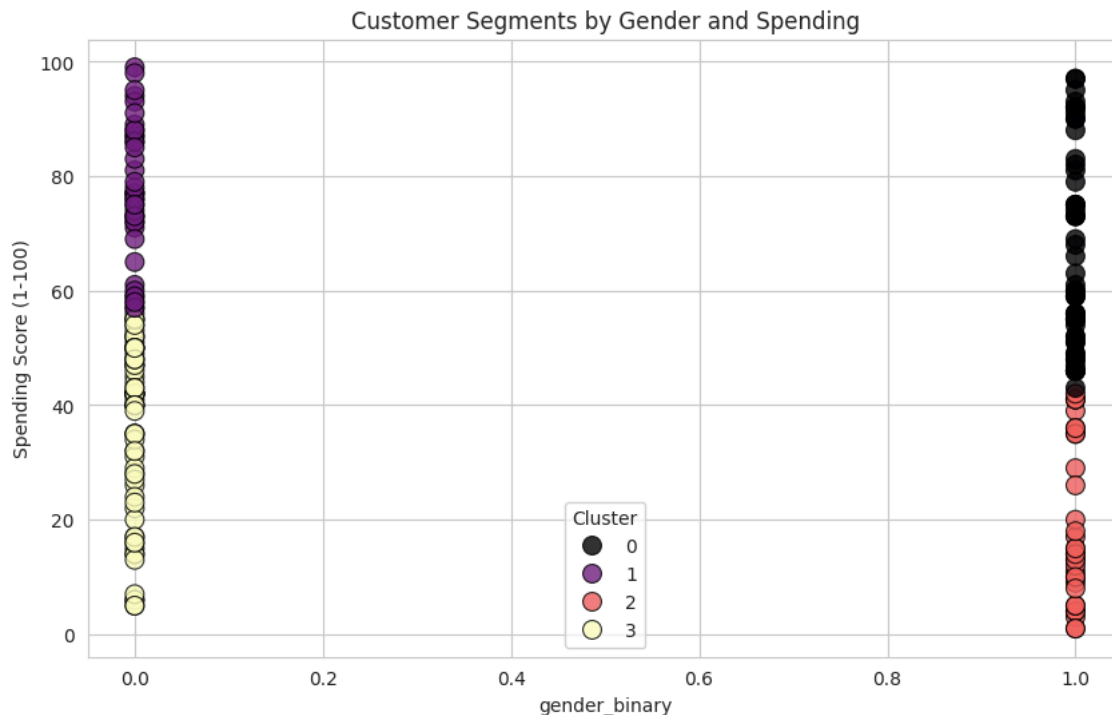
# 2. Implement the Elbow Method
wcss2 = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X3_scaled)
    wcss2.append(kmeans.inertia_)

# 3. Plot the Elbow Curve
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss2, marker='o', linestyle='--')
plt.title('Elbow Method for Gender-Spending Segmentation')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS (Inertia)')
plt.xticks(range(1, 11))
plt.show()
```



```
kmeans3 = KMeans(n_clusters=4, init='k-means++', random_state=42, n_init=10)
df['Gender_Cluster'] = kmeans3.fit_predict(X3_scaled)
```

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='gender_binary', y='Spending Score (1-100)',
                hue='Gender_Cluster', palette='magma', s=100, alpha=0.8, edgecolor='black')
plt.title('Customer Segments by Gender and Spending')
plt.legend(title='Cluster')
plt.show()
```

Interpreting the Gender-Based Segments

This model gives us a different perspective:

- **Cluster 1 : High-Spending Females:** These are the most active or highest-spending female customers.
- **Cluster 3 Budget-Conscious Females:** This group represents females with more conservative spending habits.
- **Cluster 0 High-Spending Males:** This group covers a wide range but skews toward the top half of male spenders.
- **Cluster 2 Budget-Conscious Males:** These are the lower-spending male customers.

2. Apply Feature Engineering for Clustering: Create a new feature using existing data (Feature Engineering). Then, explore if clustering can be effectively performed using this newly engineered feature in combination with one or more existing features. Document the feature engineering process, the chosen features for clustering, the determination of the optimal number of clusters (e.g., using the Elbow Method), and the resulting cluster visualization and interpretation.

```
df.head()
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income_Cluster	Age_Cluster	gender_binary	Gender_Cluster
0	Male	19	15	39	4	0	1	2
1	Male	21	15	81	2	1	1	0
2	Female	20	16	6	4	0	0	3
3	Female	23	16	77	2	1	0	1

```
df['Spending_Income_Ratio'] = df['Spending Score (1-100)'] / df['Annual Income (k$)']
df.head()
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income_Cluster	Age_Cluster	gender_binary	Gender_Cluster	Spending_Income_Ratio
0	Male	19	15	39	4	0	1	2	2.600000
1	Male	21	15	81	2	1	1	0	5.400000
2	Female	20	16	6	4	0	0	3	0.375000
3	Female	23	16	77	2	1	0	1	4.812500

```
X4 = df[['Age', 'Spending_Income_Ratio']]
scaled4 = StandardScaler()
```

```
X4_scaled = scaled4.fit_transform(X4)
```

```
wcss4 = []  
for i in range(1, 11):  
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)  
    kmeans.fit(X4_scaled)  
    wcss4.append(kmeans.inertia_)  
  
plt.figure(figsize=(10, 6))  
plt.plot(range(1, 11), wcss4, marker='o', linestyle='--')  
plt.title('Elbow Method')  
plt.xlabel('Number of Clusters (k)')  
plt.ylabel('WCSS')  
plt.show()
```

